

# Probability and Statistics

## Formula Book

GTU: BE03000251    Semester 3

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## 1 Unit 1: Basic Probability

### 1.1 Basic concepts

- **Experiment, Outcome, Sample space  $S$ .**
- **Event:** subset  $A \subseteq S$ .
- **Probability measure  $P$**  satisfies:  $0 \leq P(A) \leq 1$ ,  $P(S) = 1$ , countable additivity.

### 1.2 Axioms and simple results

$$P(\emptyset) = 0, \quad P(S) = 1.$$

$$P(A^c) = 1 - P(A).$$

### 1.3 Addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

For mutually exclusive events ( $A \cap B = \emptyset$ ):  $P(A \cup B) = P(A) + P(B)$ .

### 1.4 Conditional probability and independence

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

**Multiplication rule:**

$$P(A \cap B) = P(A | B)P(B) = P(B | A)P(A).$$

Events  $A$  and  $B$  are independent if

$$P(A \cap B) = P(A)P(B).$$

For  $n$  events mutual independence requires all finite intersections equal product of probabilities.

### 1.5 Total probability & Bayes' theorem

If  $\{A_i\}$  partition  $S$  with  $P(A_i) > 0$ , then for any event  $B$ ,

$$P(B) = \sum_i P(B | A_i)P(A_i).$$

Bayes' theorem:

$$P(A_k | B) = \frac{P(B | A_k)P(A_k)}{\sum_i P(B | A_i)P(A_i)}.$$

### 1.6 Bernoulli trials and binomial coefficient

Bernoulli trial: two outcomes (S/F). For  $n$  independent trials with success probability  $p$ ,

$$P(\text{exactly } r \text{ successes}) = \binom{n}{r} p^r (1-p)^{n-r}, \quad r = 0, 1, \dots, n.$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

### 1.7 Random variables, PMF, PDF, CDF

A **random variable**  $X$  is a real-valued function on  $S$ .

**Discrete:** Probability mass function (PMF)

$$p_X(x) = P(X = x), \quad \sum_x p_X(x) = 1.$$

**Continuous:** Probability density function (PDF)

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

Cumulative distribution function (CDF)

$$F_X(x) = P(X \leq x) = \begin{cases} \int_{-\infty}^x f_X(t) dt & \text{(continuous)} \\ \sum_{t \leq x} p_X(t) & \text{(discrete)} \end{cases}$$

Properties:  $\lim_{x \rightarrow -\infty} F(x) = 0$ ,  $\lim_{x \rightarrow \infty} F(x) = 1$ , non-decreasing, right-continuous.

## 1.8 Expectation, variance and standard deviation

**Expectation:**

$$E[X] = \begin{cases} \sum_x x p_X(x), & \text{discrete} \\ \int_{-\infty}^{\infty} x f_X(x) dx, & \text{continuous} \end{cases}$$

**Variance:**

$$\text{Var}(X) = \mu_2 = E[(X - \mu)^2] = E[X^2] - (E[X])^2.$$

**Standard deviation:**  $\sigma = \sqrt{\text{Var}(X)}$ .

## 2 Unit 2: Special Probability Distributions

### 2.1 Binomial distribution

$X \sim \text{Bin}(n, p)$

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}, \quad r = 0, 1, \dots, n.$$

Mean and variance:

$$E[X] = np, \quad \text{Var}(X) = np(1-p).$$

### 2.2 Poisson distribution

$X \sim \text{Pois}(\lambda)$

$$P(X = r) = \frac{e^{-\lambda} \lambda^r}{r!}, \quad r = 0, 1, 2, \dots$$

Mean and variance:

$$E[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

Recurrence:

$$\frac{P(X = r+1)}{P(X = r)} = \frac{\lambda}{r+1}.$$

Poisson as approximation to Binomial when  $n$  large,  $p$  small,  $np = \lambda$  fixed.

## 2.3 Normal distribution

$$X \sim N(\mu, \sigma^2)$$

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

Mean =  $\mu$ , variance =  $\sigma^2$ . Standard normal  $Z \sim N(0, 1)$ ;  $Z = (X - \mu)/\sigma$ .

## 2.4 Exponential distribution

$$X \sim \text{Exp}(\lambda), \lambda > 0$$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Mean =  $1/\lambda$ , variance =  $1/\lambda^2$ , median =  $(\ln 2)/\lambda$ .

## 2.5 Gamma distribution

$X \sim \Gamma(k, \lambda)$  with shape  $k$  (integer) and rate  $\lambda$ :

$$f_X(x) = \frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-\lambda x}, \quad x > 0.$$

Mean =  $k/\lambda$ , variance =  $k/\lambda^2$ .

# 3 Unit 3: Basic Statistics (Grouped and Ungrouped Data)

This section includes extensive formulae for **ungrouped (raw) data** and **grouped (frequency) data**. Both are given with formulas and short notes.

## 3.1 Ungrouped data (individual observations)

Suppose observations  $x_1, x_2, \dots, x_n$ .

**Sample size:**  $n$ .

**Arithmetic mean (sample):**

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

**Population mean (if population known):**

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i.$$

**Median:**

If  $n$  odd: median =  $x_{\frac{n+1}{2}}$ .

If  $n$  even: median =  $\frac{x_{n/2} + x_{n/2+1}}{2}$ .

**Mode:** value with highest frequency in the data (for unique mode).

**Range:**  $R = \max(x_i) - \min(x_i)$ .

**Sample variance (unbiased):**

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

**Population variance:**

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2.$$

**Raw moment (about origin):**

$$m'_r = \frac{1}{n} \sum_{i=1}^n x_i^r.$$

**Central moment (about mean):**

$$m_r = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^r.$$

**Coefficient of variation (CV):**

$$CV = \frac{s}{\bar{x}} \times 100\% \quad (\text{for positive mean}).$$

### 3.2 Grouped Data (Frequency Distribution)

Let classes have midpoints  $x_i$ , class width  $h$  (if equal), frequencies  $f_i$ , and total frequency  $N = \sum f_i$ .

—

**Arithmetic Mean (Grouped):**

$$\bar{x} = \frac{1}{N} \sum_i f_i x_i.$$

(If using assumed mean  $a$  and deviations  $d_i = x_i - a$ )

$$\bar{x} = a + \frac{\sum_i f_i d_i}{N}.$$

(If coding with unit  $c$ )

$$\bar{x} = a + c \cdot \frac{\sum_i f_i u_i}{N}, \quad \text{where } u_i = \frac{x_i - a}{c}.$$

### Median for Grouped Data (Continuous Approximation):

If the median class is the class whose cumulative frequency  $\geq \frac{N}{2}$ , then

$$\text{Median} = L + \frac{\frac{N}{2} - F}{f_m} \times h,$$

where  $L$  = lower boundary of the median class,  $F$  = cumulative frequency before the median class,  $f_m$  = frequency of the median class, and  $h$  = class width.

### Mode for Grouped Data (Using Formula):

If the modal class has lower boundary  $L$ , frequency  $f_m$ , preceding class frequency  $f_{m-1}$ , and succeeding class frequency  $f_{m+1}$ , then

$$\text{Mode} = L + \frac{f_m - f_{m-1}}{(2f_m - f_{m-1} - f_{m+1})} \times h.$$

### Variance (Grouped):

$$\sigma^2 = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^2 \quad (\text{population form}).$$

### Sample Variance (Unbiased):

$$s^2 = \frac{1}{N-1} \sum_i f_i (x_i - \bar{x})^2.$$

(Use  $x_i$  as class midpoints.)

### Moments for Grouped Data:

#### (i) Moments about Origin:

$$m'_r = \frac{1}{N} \sum_i f_i x_i^r.$$

These are called the *raw moments* or *moments about the origin*.

#### (ii) Moments about Mean:

$$m_r = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^r.$$

These are the *central moments*, which measure dispersion and shape of the distribution.

**(iii) Moments about Assumed Mean:**

If  $a$  is the assumed mean and  $d_i = x_i - a$ , then

$$m_r'' = \frac{1}{N} \sum_i f_i d_i^r.$$

The relationship between moments about the assumed mean and moments about the actual mean is:

$$m_1' = a + m_1'',$$

$$m_2 = m_2'' - (m_1'')^2,$$

$$m_3 = m_3'' - 3m_2''m_1'' + 2(m_1'')^3,$$

$$m_4 = m_4'' - 4m_3''m_1'' + 6m_2''(m_1'')^2 - 3(m_1'')^4.$$

**Notes:**

- Moments about the mean are independent of change of origin.
- Moments about the origin depend on the choice of origin.
- Central moments ( $m_2, m_3, m_4$ ) are used to compute variance, skewness, and kurtosis respectively.

**3.3 Other statistics for grouped data**

**Percentiles / Quartiles:**  $P_k$  (kth percentile) estimated from cumulative frequencies:

$$P_k = L + \frac{\frac{k}{100}N - F}{f_c} \times h,$$

where  $f_c$  frequency of the class containing the percentile.

**3.4 Measures of Skewness and Kurtosis (Ungrouped and Grouped Data)**

Let the  $k$ -th **central moment** about the mean be denoted by  $\mu_k$ .

**(A) For Ungrouped Data:**

For a set of  $n$  observations  $x_1, x_2, \dots, x_n$  with mean  $\bar{x}$ ,

$$\mu_k = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^k$$

**Coefficient of Skewness (Karl Pearson's moment coefficient):**

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$



**Coefficient of Kurtosis:**

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

—

**(B) For Grouped Data:**

For a frequency distribution with class mid-points  $x_j$  and frequencies  $f_j$  ( $j = 1, 2, \dots, k$ ), the total frequency is

$$N = \sum f_j,$$

and the mean is

$$\bar{x} = \frac{\sum f_j x_j}{N}.$$

Then, the central moments are given by

$$\mu_k = \frac{\sum f_j (x_j - \bar{x})^k}{N}.$$

Hence,

**Coefficient of Skewness:**

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{\left( \frac{\sum f_j (x_j - \bar{x})^3}{N} \right)^2}{\left( \frac{\sum f_j (x_j - \bar{x})^2}{N} \right)^3}.$$

**Coefficient of Kurtosis:**

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\frac{\sum f_j (x_j - \bar{x})^4}{N}}{\left( \frac{\sum f_j (x_j - \bar{x})^2}{N} \right)^2}.$$

—

**Interpretation:**

- If  $\sqrt{\beta_1} = 0$ , the distribution is symmetric. - If  $\sqrt{\beta_1} > 0$ , the distribution is positively skewed (right-skewed). - If  $\sqrt{\beta_1} < 0$ , the distribution is negatively skewed (left-skewed).  
 - If  $\beta_2 = 3$ , the distribution is mesokurtic (normal). - If  $\beta_2 > 3$ , it is leptokurtic (more peaked). - If  $\beta_2 < 3$ , it is platykurtic (flatter).

### 3.5 Grouped vs. Ungrouped quick formulas (summary)

| Ungrouped                        | Grouped                              |
|----------------------------------|--------------------------------------|
| $\bar{x} = \frac{1}{n} \sum x_i$ | $\bar{x} = \frac{1}{N} \sum f_i x_i$ |

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

Median from sorted list

Mode = most frequent value

$$s^2 = \frac{1}{N-1} \sum f_i (x_i - \bar{x})^2$$

$$\text{Median} = L + \frac{N/2 - F}{f_m} h$$

$$\text{Mode} = L + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} h$$

### 3.6 Expectation, Moments and Variance of a Probability Distribution

#### 1. Expectation (Mathematical Expectation):

For a discrete random variable  $X$  taking values  $x_1, x_2, \dots, x_n$  with corresponding probabilities  $p_1, p_2, \dots, p_n$  such that  $\sum p_i = 1$ , the **expectation** or **expected value** of  $X$  is defined as:

$$E(X) = \sum_{i=1}^n x_i p_i.$$

For a continuous random variable  $X$  with probability density function (p.d.f.)  $f(x)$ ,

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx.$$

#### 2. Moments:

The  $r$ -th **raw moment** (moment about origin) is defined as:

$$\mu'_r = E(X^r) = \sum x_i^r p_i \quad \text{or} \quad \int_{-\infty}^{\infty} x^r f(x) dx.$$

The  $r$ -th **central moment** (moment about the mean) is defined as:

$$\mu_r = E[(X - \mu)^r] = \sum (x_i - \mu)^r p_i \quad \text{or} \quad \int_{-\infty}^{\infty} (x - \mu)^r f(x) dx,$$

where  $\mu = E(X)$ .

#### 3. Variance:

The **variance** of a random variable  $X$  measures the dispersion of its values about the mean, and is defined as:

$$\text{Var}(X) = E[(X - \mu)^2] = \mu_2.$$

It can also be written in terms of raw moments as:

$$\text{Var}(X) = E(X^2) - [E(X)]^2.$$

The **standard deviation** is the positive square root of variance:

$$\sigma = \sqrt{\text{Var}(X)}.$$

**4. Important Properties of Expectation:**

If  $a$  and  $b$  are constants, and  $X$  and  $Y$  are random variables:

1.  $E(aX + b) = aE(X) + b$
2.  $E(X + Y) = E(X) + E(Y)$
3.  $E(aX + bY) = aE(X) + bE(Y)$
4. If  $X$  and  $Y$  are independent, then  $E(XY) = E(X)E(Y)$

**5. Properties of Variance:**

1.  $\text{Var}(aX + b) = a^2 \text{Var}(X)$
2.  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$
3. If  $X$  and  $Y$  are independent, then  $\text{Cov}(X, Y) = 0$ , and hence  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

**6. Relation Between Moments:**

$$\begin{aligned}\mu_1 &= 0, \\ \mu_2 &= \mu'_2 - (\mu'_1)^2, \\ \mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3, \\ \mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4.\end{aligned}$$

These relationships are useful in converting raw moments into central moments.

**7. Special Note:**

For a symmetric distribution about the mean,

$$\mu_3 = 0, \quad \text{and all odd central moments vanish.}$$

## 4 Correlation and Regression

### 4.1 Correlation

Given paired data  $(x_i, y_i), i = 1, \dots, n$ .

**Sample covariance:**

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n\bar{x}\bar{y}.$$

**Pearson correlation coefficient:**

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}.$$

Properties:  $-1 \leq r \leq 1$ .

## 4.2 Regression lines

**Regression of  $y$  on  $x$ :**

$$\hat{y} = a + bx, \quad b_{yx} = \frac{S_{xy}}{S_{xx}}, \quad a = \bar{y} - b\bar{x},$$

where  $S_{xx} = \sum (x_i - \bar{x})^2$ .

**Regression of  $x$  on  $y$ :**

$$\hat{x} = a' + b'y, \quad b_{xy} = \frac{S_{xy}}{S_{yy}}.$$

**Relation with  $r$ :**

$$b_{yx} = r \frac{S_y}{S_x}, \quad b_{xy} = r \frac{S_x}{S_y}.$$

Product of slopes:  $b_{yx}b_{xy} = r^2$ .

## 4.3 Rank Correlation (Spearman)

If ranks of  $n$  individuals in two characteristics are  $R_i$  and  $S_i$ , and the difference between ranks is

$$d_i = R_i - S_i,$$

then the **Spearman's Rank Correlation Coefficient** is given by:

$$R_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}.$$

**For Repeated (Tied) Ranks:**

When ranks are repeated (tied), correction terms are added for each set of tied ranks.

Let: - For the first variable, ties occur in groups of sizes  $t_1, t_2, t_3, \dots$  - For the second variable, ties occur in groups of sizes  $u_1, u_2, u_3, \dots$

Then, the corrected formula is:

$$R_s = 1 - \frac{6 \left[ \sum d_i^2 + \frac{1}{12} \left( \sum (t_j^3 - t_j) + \sum (u_k^3 - u_k) \right) \right]}{n(n^2 - 1)}.$$

**Notes:** -  $R_s$  lies between  $-1$  and  $+1$ . -  $R_s = +1$  indicates perfect positive correlation of ranks. -  $R_s = -1$  indicates perfect negative correlation. -  $R_s = 0$  indicates no correlation between the rankings.

## 5 Unit 4: Applied Statistics — Hypothesis Testing

### 5.1 General procedure

1. State null hypothesis  $H_0$  and alternative  $H_1$ .

2. Choose significance level  $\alpha$  (commonly 0.05 or 0.01).
3. Determine test statistic and its distribution under  $H_0$ .
4. Compute the observed value and p-value or compare with critical value.
5. Make decision: reject or fail to reject  $H_0$ .

## 5.2 Large-sample tests (Z-tests)

**One-sample mean (known  $\sigma$ ):**

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1) \text{ under } H_0.$$

**Two-sample mean (known variances):**

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}.$$

**Test for proportion:** For sample proportion  $\hat{p}$  and null  $p_0$ ,

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}.$$

Two-sample proportion difference:

$$Z = \frac{p_1 - p_2}{\sqrt{P(1 - P) \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}}, \quad P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}.$$

## 5.3 Small-sample tests (Student's t)

If population is normal and  $\sigma$  unknown, use  $t$  with  $n - 1$  df.

**One-sample t:**

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}.$$

**Two-sample t (pooled variance, equal variances):**

$$S_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \quad t = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

with  $n_1 + n_2 - 2$  degrees of freedom.

## 5.4 t-test for correlation coefficient

Testing  $H_0 : \rho = 0$  with sample correlation  $r$  ( $n$  pairs):

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}.$$

## 5.5 F-test for ratio of variances

For independent samples,

$$F = \frac{s_1^2}{s_2^2} \sim F_{n_1-1, n_2-1} \quad (\text{under } H_0 : \sigma_1^2 = \sigma_2^2).$$

## 5.6 Chi-square test (goodness-of-fit)

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi_{k-c-1}^2,$$

where  $c$  = number of parameters estimated from data; compare with critical  $\chi^2$ .

## 5.7 Chi-square test (contingency / independence)

Construct contingency table; compute expected counts under independence and use same  $\chi^2$  formula with degrees of freedom  $(r-1)(c-1)$ .

# 6 Unit 5: Curve Fitting by Least Squares

## 6.1 Fitting straight line $y = a + bx$

Normal equations:

$$\sum y = na + b \sum x, \quad \sum xy = a \sum x + b \sum x^2.$$

Solve to get

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}, \quad a = \bar{y} - b\bar{x}.$$

## 6.2 Fitting second degree parabola $y = a + bx + cx^2$

Normal equations:

$$\begin{cases} \sum y = na + b \sum x + c \sum x^2, \\ \sum xy = a \sum x + b \sum x^2 + c \sum x^3, \\ \sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4. \end{cases}$$

Solve linear system for  $a, b, c$ .

## 6.3 Fitting exponential curve $y = ae^{bx}$

Take logarithms:

$$\ln y = \ln a + bx.$$

Let  $Y = \ln y$ ,  $A = \ln a$ . Then  $Y = A + bx$ .

Normal equations:

$$\sum Y = nA + b \sum x, \quad \sum xY = A \sum x + b \sum x^2.$$

Solve for  $A$  and  $b$ , then  $a = e^A$ .

## 6.4 Fitting power curve $y = ax^b$

Take logarithms:

$$\ln y = \ln a + b \ln x.$$

Let  $Y = \ln y$ ,  $X = \ln x$ , and  $A = \ln a$ . Then  $Y = A + bX$ .

Normal equations:

$$\sum Y = nA + b \sum X, \quad \sum XY = A \sum X + b \sum X^2.$$

Solve for  $A$  and  $b$ , then  $a = e^A$ .

## 6.5 Fitting geometric (exponential-type) curve $y = ab^x$

Take logarithms:

$$\ln y = \ln a + x \ln b.$$

Let  $Y = \ln y$ ,  $A = \ln a$ , and  $B = \ln b$ . Then  $Y = A + Bx$ .

Normal equations:

$$\sum Y = nA + B \sum x, \quad \sum xY = A \sum x + B \sum x^2.$$

Solve for  $A$  and  $B$ , then  $a = e^A$ ,  $b = e^B$ .

# 7 Additional Important Formulae and Tables

## 7.1 Common expectations

$$E[aX + b] = aE[X] + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X).$$

If  $X, Y$  independent,

$$E[XY] = E[X]E[Y], \quad \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

## 7.2 Sampling distributions (brief)

For  $X_i$  i.i.d. with mean  $\mu$  and variance  $\sigma^2$ ,

$$\bar{X} \text{ has } E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

By CLT, for large  $n$ ,  $\bar{X} \approx N(\mu, \sigma^2/n)$ .

## 7.3 Probability limits and inequalities

**Markov:** For nonnegative  $X$ ,

$$P(X \geq a) \leq \frac{E[X]}{a}, \quad a > 0.$$

**Chebyshev:**

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

## 8 Appendices

### 8.1 Appendix A: Table of common distributions summary

| Distribution              | PMF / PDF                                                | Mean, Variance           |
|---------------------------|----------------------------------------------------------|--------------------------|
| Binomial( $n, p$ )        | $\binom{n}{r} p^r (1-p)^{n-r}$                           | $np, np(1-p)$            |
| Poisson( $\lambda$ )      | $e^{-\lambda} \lambda^r / r!$                            | $\lambda, \lambda$       |
| Normal( $\mu, \sigma^2$ ) | $\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$ | $\mu, \sigma^2$          |
| Exponential( $\lambda$ )  | $\lambda e^{-\lambda x}, x \geq 0$                       | $1/\lambda, 1/\lambda^2$ |
| Gamma( $k, \lambda$ )     | $\frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-\lambda x}$     | $k/\lambda, k/\lambda^2$ |

### 8.2 Appendix B: Quick reference for grouped data formulas

- Mean:  $\bar{x} = \frac{1}{N} \sum f_i x_i$ .
- Median:  $L + \frac{N/2 - F}{f_m} h$ .
- Mode:  $L + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} h$ .
- Variance:  $\frac{1}{N} \sum f_i (x_i - \bar{x})^2$  (population).